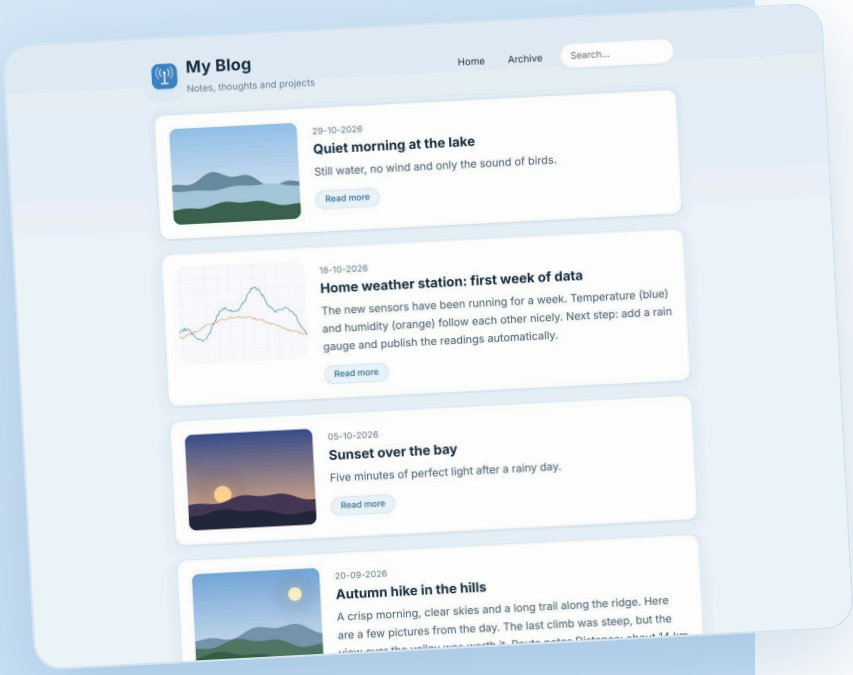
NXBlogger

Installation, Update, Backup and User Guide

Python · Flask | nginx · HTTPS | Let's Encrypt | Paste screenshots | WordPress import | SQLite



Version 1.0.1 · 03-11-2026

https://github.com/oh2gax/nxblogger

Contents

1	Overview	What NXBlogger is and how it works
2	Installation	Server, DNS, firewall, installer, HTTPS certificate
3	Writing and managing posts	Login, editor, images, publishing, settings
4	What visitors see	Front page, posts, archive, tags, search, RSS
5	Importing from WordPress	Export file, images, web and command-line import
6	Updating	update.sh, automatic updates, rollback
7	Backup, export and restore	Where data lives, backups, SQLite, restore
8	Configuration reference	Installer options and settings file
9	Troubleshooting	Common problems and where to look
10	Command cheat sheet	All commands on one page

Examples in this guide use the domain `blog.example.com`, the admin user `admin` and the program folder `/opt/nxblogger`. Replace them with your own values.

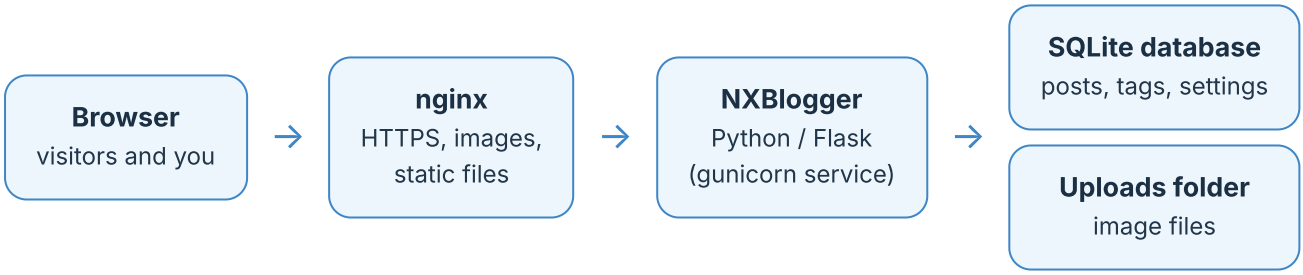
1 Overview

NXBlogger is a small, calm blog publishing system. It is built for one author who wants to write posts with text and images, without plugins, comments or tracking. Images can be added by simply **pasting a screenshot** into the editor. An existing WordPress blog can be imported, including its images, dates and tags.

Features

- Light blue, simple layout with rounded buttons that also works on phones.
- Maintenance pages to write, edit, publish, unpublish and delete posts.
- Images by clipboard paste (Ctrl+V), drag & drop or file picker.
- Drafts, publish dates, tags, archive, search and an RSS feed.
- WordPress import from the standard export file, with old WordPress links redirected to the new addresses.
- HTTPS with a free Let's Encrypt certificate that renews automatically.
- One-command installation, one-command updates from GitHub, and daily database backups.
- One admin user. The password is stored only as a salted hash, never in plain text.

How it works



nginx receives all requests on the HTTPS port and passes page requests to the NXBlogger service, which listens only on the server itself (`127.0.0.1:8801`). Images and style files are sent directly by nginx. All posts are stored in one SQLite database file, and images are ordinary files in the uploads folder.

Requirements

Item	Details
Server	Ubuntu 24.04 LTS, for example on AWS EC2 or AWS Lightsail. 1 GB of memory is enough.
Domain name	A DNS name for the blog, e.g. `blog.example.com` , pointing to the server's IP address.
Open ports	TCP 80 (certificate validation and redirect) and the HTTPS port: 443, or any other port such as 8800.
Access	SSH login with a user that can use `sudo` .

2 Installation

2.1 Prepare the server, DNS and firewall

1. **Server:** create an Ubuntu 24.04 LTS instance and give it a fixed public IP address (AWS: *Elastic IP*; Lightsail: *Static IP*).
2. **DNS:** create an **A record** `blog.example.com → server IP`. If you move an existing blog, lower the record's TTL to about 300 seconds a day in advance, so the change spreads quickly.
3. **Firewall:** allow inbound **TCP 80** and your HTTPS port (**443**, or e.g. 8800). AWS EC2: security group. Lightsail: instance → *Networking* → IPv4 and IPv6 firewall. Allow SSH (TCP 22) preferably only from your own IP address.
4. **Update Ubuntu** and reboot if asked:

```
sudo apt-get update && sudo apt-get -y upgrade
sudo reboot
```

2.2 Get the code

```
sudo apt-get install -y git
sudo git clone https://github.com/oh2gax/nxblogger.git /opt/nxblogger
```

Home folder instead of /opt? You can also clone as your own user, without `sudo`: `git clone https://github.com/oh2gax/nxblogger.git ~/nxblogger`. The service then runs as your user, like other programs in your home folder. Use `~/nxblogger` instead of `/opt/nxblogger` in all commands.

2.3 Run the installer

```
sudo bash /opt/nxblogger/install.sh --domain blog.example.com --port 443 --title "My Blog" --
admin-user admin
```

The installer asks for the **admin password** (at least 8 characters). Then it:

1. installs the system packages (Python, nginx, certbot, sqlite3 and others),
2. creates the service user and the data folders under `/var/lib/nxblogger`,
3. writes the settings file `/etc/nxblogger/nxblogger.env` with a random secret key,
4. creates the Python environment and the database,
5. configures nginx and starts the `nxblogger` service,
6. requests a Let's Encrypt certificate and sets up automatic renewal,
7. sets up a daily database backup.

When it finishes, open **https://blog.example.com/**. The maintenance pages are at **https://blog.example.com/admin/**, and there is also a small *Login* link at the bottom of every page.

Safe to run again. The installer keeps your posts, images, password and settings. Run it again to change an option (e.g. `--port 8800`), to retry the certificate, or to turn automatic updates on.

Installer option	Default	Meaning
`--domain NAME`	required	DNS name of the blog.
`--port N`	8800	Public HTTPS port. Use 443 for addresses without a port number.
`--title TEXT`	My Blog	Blog title. It can be changed later in Settings.
`--admin-user NAME`	admin	Name of the admin user.
`--admin-password PW`	asked	Initial admin password (otherwise asked).
`--email ADDR`	–	Optional e-mail registered with Let's Encrypt.
`--app-port N`	8801	Internal port of the service (127.0.0.1 only).
`--self-signed`	–	Skip Let's Encrypt and use a self-signed certificate.
`--auto-update` / `--no-auto-update`	off	Nightly automatic update from GitHub.

2.4 The HTTPS certificate

- Let's Encrypt checks the domain over **port 80**, so the DNS name must already point to the server and port 80 must be open.
- If that is not the case yet, the installer prints a warning and the blog runs with a temporary self-signed certificate (browsers show a warning). Fix the cause and run the installer again.
- Let's Encrypt allows only a few failed attempts per hour. Check DNS first, e.g. with `dig +short @1.1.1.1 blog.example.com`.
- The certificate renews automatically. Test the renewal with `sudo certbot renew --dry-run`.

2.5 Servers that already run other websites

NXBlogger adds one nginx site and leaves the others alone. Several sites can share port 443, because nginx tells them apart by name. The installer stops with a clear message if another site already uses the same domain name, or if the internal port 8801 is taken (choose another one with `--app-port`). nginx restarts once during installation, so other sites pause for about a second.

2.6 Check that everything works

```
systemctl status nxblogger --no-pager        # should show "active (running)"
curl -s https://blog.example.com/healthz      # should print {"status": "ok"}
```

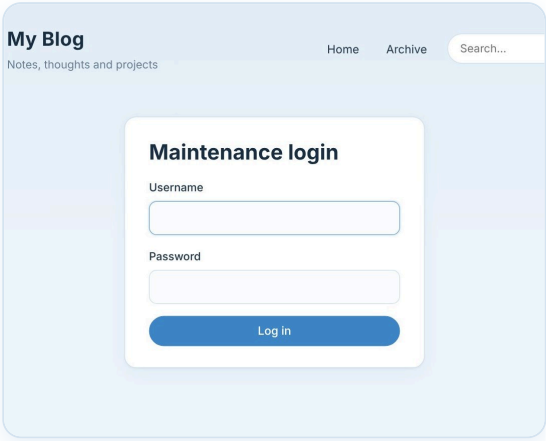
3 Writing and managing posts

3.1 Logging in

Open **https://blog.example.com/admin/** or click *Login* at the bottom of any page, and log in with the admin user and password.

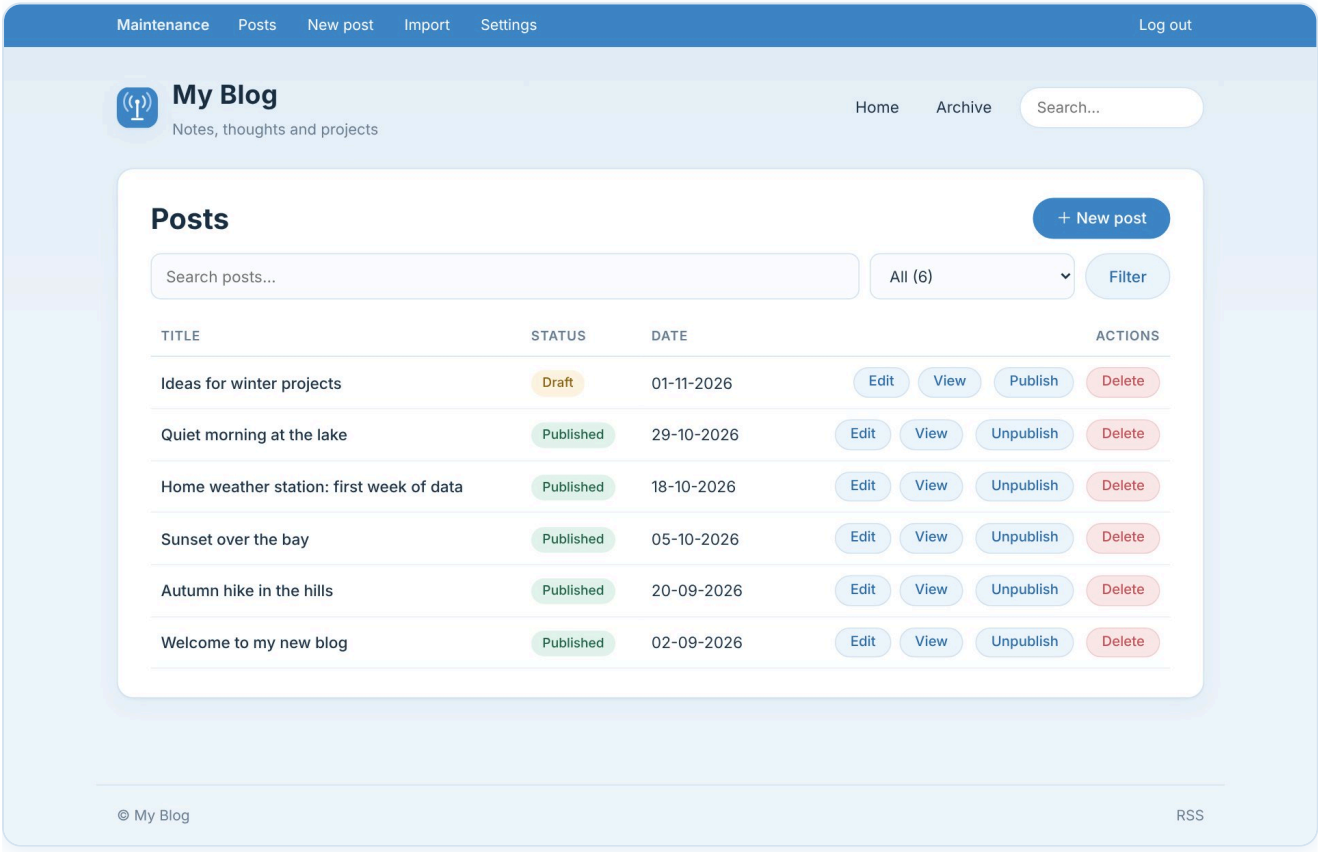
When you are logged in, a blue **Maintenance** bar appears at the top of every page with links to *Posts*, *New post*, *Import*, *Settings* and *Log out*.

> **Security:** after 5 wrong passwords from the same address, login is blocked for 15 minutes. A login stays valid for 12 hours.



Login page

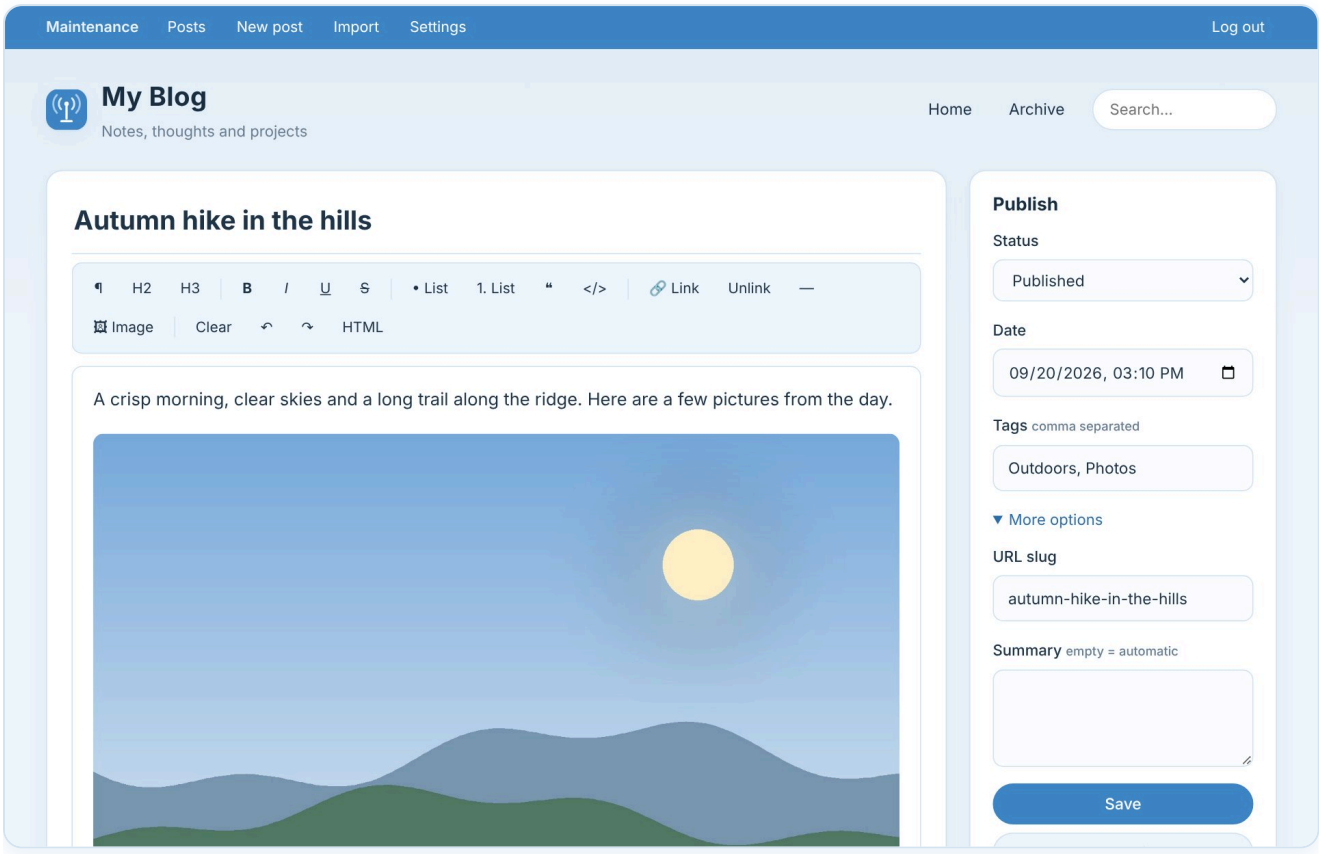
3.2 The post list



Maintenance → Posts: search, filter by status, and quick actions for each post

Maintenance → Posts lists all posts, newest first. Use the search box and the status filter (All / Published / Drafts) to find posts. Each row has buttons to **Edit**, **View**, **Publish/Unpublish** and **Delete**. Deleting asks for confirmation and cannot be undone. The images of a deleted post stay in the uploads folder.

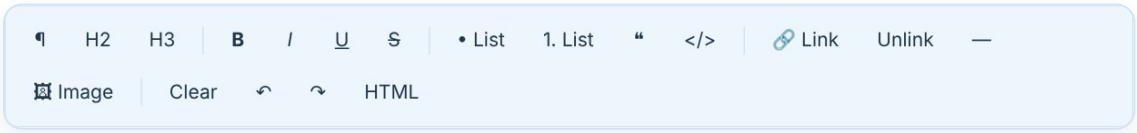
3.3 Writing a post



The editor: title, formatting toolbar, text area and the Publish box on the right

1. Click **New post** in the Maintenance bar.
2. Type the **title**, then click in the text area and write. The text appears just as it will look in the blog.
3. Add images (see 3.4) and formatting from the toolbar.
4. In the **Publish** box choose **Draft** or **Published**, and click **Save**. **Save & view** opens the post afterwards. **Ctrl+S** also saves.

The toolbar



Button	What it does
¶ · H2 · H3	Normal paragraph, heading, subheading.
B · I · U · S	Bold, italic, underline, strikethrough (also Ctrl+B, Ctrl+I, Ctrl+U).
• List · 1. List	Bulleted or numbered list.
❝ · </>	Quote block and code block. Click again to turn it off.
Link · Unlink	Make the selected text a link, or remove a link.
—	Horizontal divider line.
Image	Choose one or more image files to insert.
Clear · ↶ · ↷	Remove formatting from the selection, undo, redo.
HTML	Show and edit the HTML source. Click again to return to the normal view.

3.4 Adding images

There are three ways to add images. Each image is uploaded immediately; it looks faded until the upload has finished.

Way	How
Paste	Take a screenshot (Windows: **Win+Shift+S**, Mac: **Cmd+Ctrl+Shift+4**) or copy an image, click in the text where it should go, and press **Ctrl+V**.
Drag & drop	Drag one or more image files from your file manager into the text.
Image button	Click **Image** in the toolbar and choose one or more files.

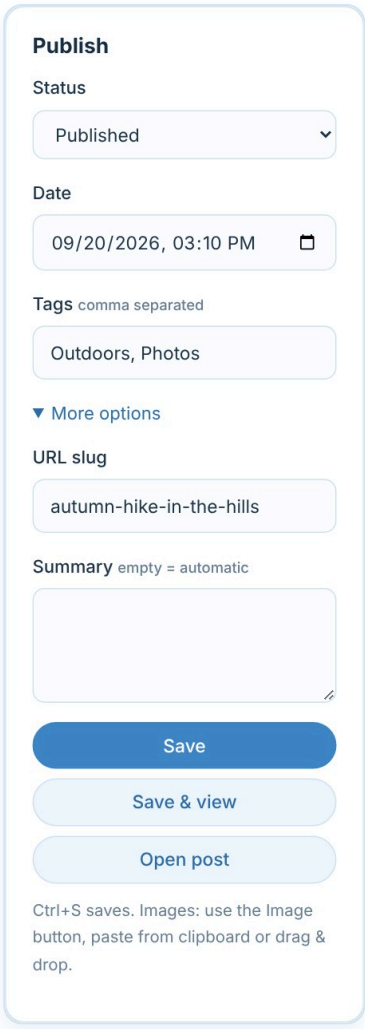
- Allowed formats: PNG, JPEG, GIF and WebP, up to 25 MB per image.
- To remove an image, click it (it gets a blue frame) and press **Delete** or **Backspace**.
- The first image of a post is shown as its thumbnail on the front page.
- Visitors can click an image to see it larger.

3.5 Publish options

Field	Meaning
Status	*Draft* is visible only to you (with a "draft" note). *Published* is visible to everyone.
Date	The publish date shown on the post. Posts are ordered by this date, so you can also back-date a post.
Tags	Keywords separated by commas, e.g. `Photos, Outdoors`. Visitors can click a tag to see all posts with it.
URL slug	The last part of the post's address (`/post/autumn-hike-in-the-hills`). It is made from the title automatically.
Summary	The short text on the front page. Leave it empty to use the beginning of the post.

URL slug and *Summary* are under **More options**.

Leaving the page? If you have unsaved changes, the browser asks before closing the editor.

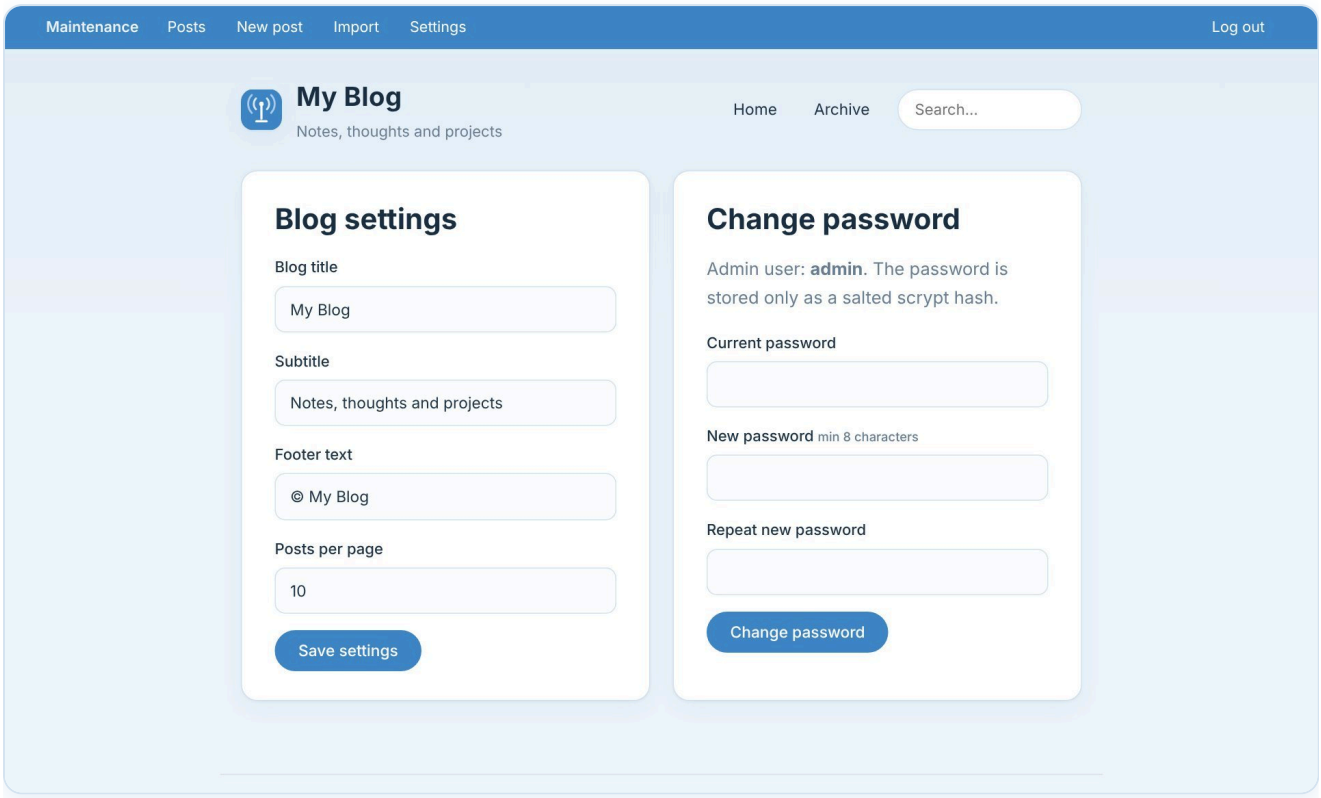


Publish box

3.6 Editing, unpublishing and deleting

- **Edit:** click the title in the post list, or **Edit post** on the post itself when you are logged in.
- **Unpublish:** set the status to *Draft*, or use **Unpublish** in the post list. The post disappears from the blog but is kept.
- **Delete: Delete** in the post list, or **Delete post** at the bottom of the editor.

3.7 Settings and password



Maintenance → Settings

- **Blog settings:** title, subtitle, footer text and the number of posts per page.
- **Change password:** enter the current password and the new one twice (at least 8 characters).
- **Forgot the password?** Reset it on the server: `sudo nxblogger-manage set-password`

4 What visitors see

Front page

The newest posts as cards with thumbnail, date, title and summary, 10 per page (adjustable in Settings), with *Newer / Older* buttons at the bottom.

Single post

The full post with its date and tags, and links to the previous and next post at the bottom. Clicking an image opens a larger view.

Archive, tags and search

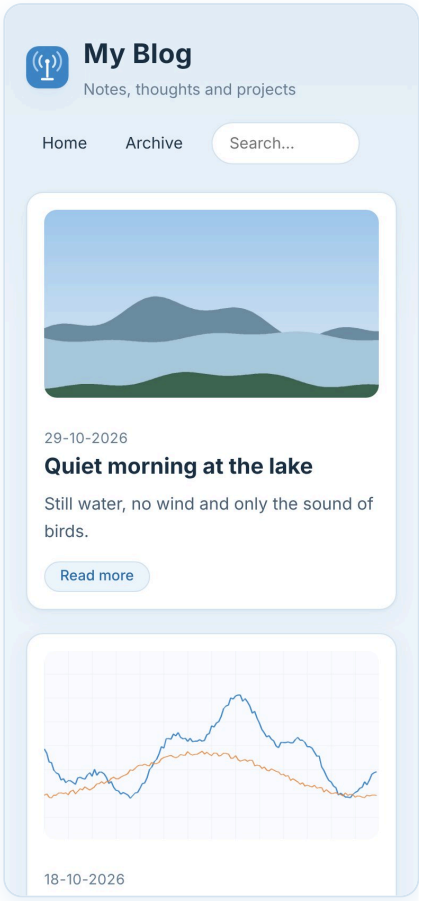
Archive lists all posts by year, with all tags at the top. A **tag** shows all posts with that tag. The **search** box in the header searches titles and text.

RSS feed

The *RSS* link in the footer (`/feed.xml`) lets readers follow the blog in a feed reader.

Phones

The layout adapts to small screens automatically.



On a phone



A single post

5 Importing from WordPress

5.1 What you need from WordPress

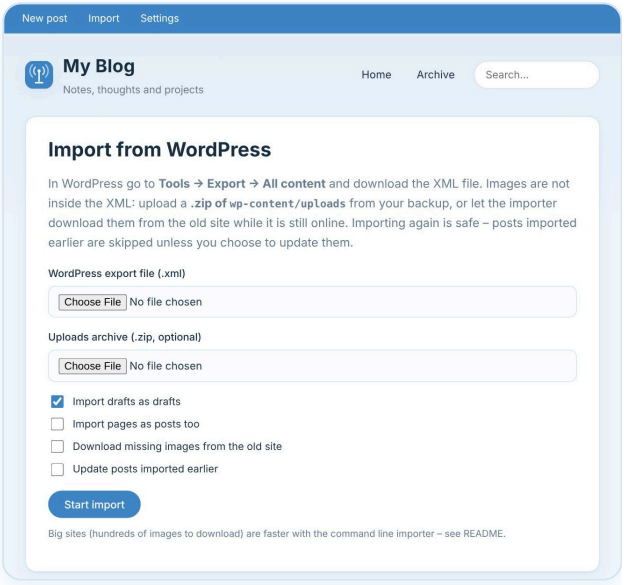
1. **The export file:** in WordPress admin, go to **Tools → Export → All content → Download Export File**. This `.xml` file contains posts, drafts, tags, categories and dates, but not the image files.
2. **The images:** the folder `wp-content/uploads` as a **.zip**. Get it from your hosting file manager, by FTP (then zip it on your PC), or from a backup plugin such as UpdraftPlus (its `…-uploads.zip`).

> **Moving the blog to a new server?** Take the export and the images **before** pointing the domain name to the new server. If you have no zip, the importer can download the images from the old site, but only while it is still online.

5.2 Import in the browser

Open **Maintenance → Import**, choose the `.xml` file and optionally the uploads `.zip`, tick the options and click **Start import**. A report shows how many posts were imported and how many images were copied.

The browser import accepts uploads up to 200 MB. For larger zips, or for downloading many images, use the command line.



Maintenance → Import

5.3 Import on the command line

Copy the files to the server (e.g. with WinSCP, or `scp export.xml uploads.zip user@server:~/`), then:

```
sudo nxblogger-manage import-wp ~/export.xml --uploads ~/uploads.zip

# without a zip, while the old site is still online:
sudo nxblogger-manage import-wp ~/export.xml --download-images
```

Option	Meaning
`--uploads FILE|DIR`	Zip file or folder containing `wp-content/uploads`.
`--download-images`	Download missing images from the old site.
`--skip-drafts`	Do not import drafts (by default they are imported as drafts).
`--include-pages`	Also import WordPress pages, as ordinary posts.
`--update-existing`	Overwrite posts that were imported earlier.

5.4 What the importer does

- Keeps titles, publish dates, drafts, tags and categories ("Uncategorized" is skipped) and featured images.
- Converts classic-editor text into paragraphs and cleans block-editor content. Captions become images with captions, galleries become image grids, and YouTube links become embedded players.
- Rewrites image links to the copied images (`/uploads/wp/…`). The report tells you if some images still point to the old site.
- Old WordPress addresses such as `/2025/08/12/post-name/` and `/?p=123` redirect to the new post addresses.
- It is safe to run again: posts imported earlier are skipped, unless you choose to update them.

After the import, delete the export files from the server; they are not needed any more.

6 Updating

New versions are published on GitHub. To update the server, run:

```
sudo bash /opt/nxblogger/update.sh
```

`update.sh` does everything in one go:

1. Checks GitHub for new commits. If there are none, it stops ("Already up to date").
2. Shows the new changes and downloads them.
3. Backs up the database.
4. Updates the Python packages and, if needed, the database structure.
5. Refreshes the nginx and service configuration.
6. Restarts the blog and checks that it answers.

To re-apply the current version, e.g. after editing the settings file, use `sudo bash /opt/nxblogger/update.sh --force`.

6.1 Automatic updates

```
sudo bash /opt/nxblogger/install.sh --auto-update       # on: every night around 04:00
sudo bash /opt/nxblogger/install.sh --no-auto-update    # off
journalctl -u nxblogger-update                           # what the last runs did
```

6.2 If something goes wrong

- If the new version does not start, `update.sh` prints the exact commands to return to the previous version.
- **"Local changes would be overwritten":** files in the program folder were edited on the server. Undo the edits with `sudo git -C /opt/nxblogger checkout -- .` and run `update.sh` again. Make changes in your own copy of the code and publish them through GitHub instead of editing on the server.

Your content is not touched by updates. Posts, images and settings live in `/var/lib/nxblogger`, separate from the program folder, and the database is backed up before every update.

7 Backup, export and restore

7.1 Where your data lives

What	Where
Posts, tags, settings, admin user	`/var/lib/nxblogger/data/nxblogger.db` (SQLite database)
Images	`/var/lib/nxblogger/uploads/`
Backups	`/var/lib/nxblogger/backups/`
Settings file	`/etc/nxblogger/nxblogger.env`
Program	`/opt/nxblogger` (or your home folder); can always be downloaded again from GitHub

A complete copy of the blog is the **database plus the uploads folder**.

7.2 Automatic backups

- **Every night at 03:30** a copy of the database is saved to `/var/lib/nxblogger/backups/`. The newest 30 are kept.
- **Before every update** an extra database copy is made. Together with the nightly copies, the newest 30 are kept.

> **These backups are on the same server.** Also download a full backup now and then, or turn on automatic server snapshots at your cloud provider (Lightsail: instance → Snapshots → *Enable automatic snapshots*).

7.3 Full backup and export

A full backup puts the database and all images into one `.tar.gz` file:

```
sudo nxblogger-manage backup --keep 5
ls -l /var/lib/nxblogger/backups/
```

The file is named like `nxblogger-full-20261103-174512.tar.gz`; `--keep 5` keeps only the five newest. The backup command copies the database safely while the blog is running. Download the file to your PC (e.g. with WinSCP) and unpack it with 7-Zip or `tar -xzf`. Inside are `nxblogger.db` and the `uploads` folder.

For a database-only backup use `sudo nxblogger-manage backup --db-only`.

7.4 Looking inside the database

The database is a normal SQLite file. On Windows, Mac or Linux the free **DB Browser for SQLite** opens it. The tables are `posts` (title, slug, HTML content, summary, status, dates), `post_tags`, `settings` and `users` (only the password hash). A quick look on the server:

```
sqlite3 /var/lib/nxblogger/data/nxblogger.db "select published_at, status, title from posts
order by published_at"
```

Work on a copy when you explore the database. Don't edit the live file while the blog is running.

7.5 Restore

To restore a full backup, on the same server or on a new server after running the installer:

```
sudo systemctl stop nxblogger
mkdir -p /tmp/restore
sudo tar -xzf nxblogger-full-YYYYMMDD-HHMMSS.tar.gz -C /tmp/restore
sudo cp /tmp/restore/nxblogger.db /var/lib/nxblogger/data/nxblogger.db
sudo rm -f /var/lib/nxblogger/data/nxblogger.db-wal /var/lib/nxblogger/data/nxblogger.db-shm
sudo cp -a /tmp/restore/uploads/. /var/lib/nxblogger/uploads/
sudo bash /opt/nxblogger/update.sh --force      # fixes file permissions and starts the blog
```

The restored database includes the admin user and password from the time of the backup.

8 Configuration reference

Most settings are given to the installer and saved in `/etc/nxblogger/nxblogger.env`. After editing that file by hand, apply it with `sudo bash /opt/nxblogger/update.sh --force`. The blog title, subtitle, footer text and posts per page are changed in **Maintenance → Settings** instead.

Setting	Default	Meaning
`NXB_DOMAIN`	–	Domain name of the blog and its certificate.
`NXB_HTTPS_PORT`	8800	Public HTTPS port served by nginx.
`NXB_APP_PORT`	8801	Internal port of the service (127.0.0.1 only).
`NXB_ADMIN_USER`	admin	Admin user name.
`NXB_BLOG_TITLE`	My Blog	Initial blog title.
`NXB_TIMEZONE`	Europe/Helsinki	Time zone used for post dates.
`NXB_MAX_UPLOAD_MB`	200	Largest upload, e.g. a WordPress import zip.
`NXB_MAX_IMAGE_MB`	25	Largest single image.
`NXB_GUNICORN_WORKERS`	2	Number of worker processes.
`NXB_SERVICE_USER`	automatic	User the service runs as.
`NXB_DATA_DIR`	/var/lib/nxblogger	Database, uploads and backups.
`NXB_SECRET_KEY`	random	Signs login cookies. Keep it secret; changing it logs you out.

9 Troubleshooting

Problem	What to check
The page does not open	Is the HTTPS port open in the firewall? Does the domain point to the server (`dig +short @1.1.1.1 blog.example.com`)? On the server: `systemctl status nxblogger` .
Works for others but not on my PC	Your DNS may still remember the old address. Windows: `ipconfig /flushdns` , then `nslookup blog.example.com` ; or wait for the DNS to update.
Certificate warning in the browser	Let's Encrypt has not succeeded yet. Check DNS and that port 80 is open, then run the installer again.
"Bad request" when saving	The page was open for a very long time or you logged out in another tab. Copy your text, reload the page and save again.
"Upload is too large"	Raise `NXB_MAX_UPLOAD_MB` or `NXB_MAX_IMAGE_MB` and run `update.sh --force` .
Forgot the admin password	`sudo nxblogger-manage set-password`
Login blocked	After 5 wrong passwords, wait 15 minutes.
update.sh: local changes	See section 6.2.
Error messages	App: `journalctl -u nxblogger -n 100` nginx: `/var/log/nginx/nxblogger.error.log`
Timers (backup, auto-update)	`systemctl list-timers 'nxblogger*'`

10 Command cheat sheet

Install and update

```
sudo git clone https://github.com/oh2gax/nxblogger.git /opt/nxblogger
sudo bash /opt/nxblogger/install.sh --domain blog.example.com --port 443 --title "My Blog" --
admin-user admin
sudo bash /opt/nxblogger/install.sh                    # re-run: retry certificate, keep
settings
sudo bash /opt/nxblogger/update.sh                     # update from GitHub
sudo bash /opt/nxblogger/update.sh --force             # re-apply settings / repair
sudo bash /opt/nxblogger/install.sh --auto-update      # nightly updates on
```

Management

```
sudo nxblogger-manage set-password                     # reset admin password
sudo nxblogger-manage import-wp ~/export.xml --uploads ~/uploads.zip
sudo nxblogger-manage backup --keep 5                  # full backup (database + images)
sudo nxblogger-manage backup --db-only                 # database only
```

Service and logs

```
systemctl status nxblogger
sudo systemctl restart nxblogger
journalctl -u nxblogger -n 100
sudo tail -50 /var/log/nginx/nxblogger.error.log
systemctl list-timers 'nxblogger*'
sudo certbot renew --dry-run
```

Addresses

`/`	Front page
`/post/<slug>`	A post
`/archive` · `/tag/<tag>` · `/?q=word`	Archive, tag, search
`/feed.xml`	RSS feed
`/admin/`	Maintenance (login)
`/healthz`	Health check for monitoring